

Enhancement Three: Databases Narrative

The artifact chosen to represent databases is my android inventory tracking application created in the java programming language. It was designed to track inventory items in an SQLite database where the user can access inventory items to either read items on a grid view, add new items by name and quantity, update item by quantity, or delete an item. Through previous enhancements the user is also able to search through the inventory alphabetically by item name and to sort by quantity high to low and low to high. I created this application at the end of 2025 for my mobile architect and programming course at Southern New Hampshire University.

I chose this artifact because it demonstrates my ability to work with an SQLite database and write advanced and secure queries that are appropriate to industry standards, Within the code I created a place holder for the query needed to search for the item based on user input, then I used selection args to define the values of the placeholder from the query. This kept the data hidden and secured the code from SQL injection attacks that could happen from exposed data. This demonstrates awareness to the security vulnerabilities presented within databases and emphasizes the importance and value of protecting data.

With the creation of the search mechanism there is heavy emphasis on the search method created in the DatabaseHelper class to showcase my competency in databases, but with this I also needed to create UI components to access this information. I created a search mechanism for the user to interact with. I created an image button that is an icon of a magnify glass to indicate to the user this is a search option available. From there I created a SearchView that is popular with Android Development that allowed the user to input the item name for searching. I applied visibility tricks (View.GONE and View.INVISIBLE) to hide features to give more room for the input field and eliminate the need for creating additional screens to the user. I utilized the current

screen the user was on and manipulated the components to disappear based on the user behavior. An example of this was when the user clicked the search icon (magnify glass) the spinner (drop down sort feature) would disappear (hide) along with disappearance of the searchIcon this allowed more room for the search view input field where hint “search item name” was visible to the user. Without hiding the visibility, the user interface would become crowded, and visibility would be restricted to the user. This demonstrated important principles to UI/UX design that kept the design clean, readable, intuitive, and provided seamless navigation to the applications features. Demonstration of skills needed for android development through implementation of the “tricks” with the use of materials and libraries available in android, application of java principles to create objects and instances of the data, effectively hiding query statements through industry standard security best practices, and development with the user in mind through user interface and user experience design principles.

I met the course outcomes I planned from Module One. All five course outcomes should be achieved with the submission of this milestone. I successfully created the search method, added the search mechanism to the UI, and wrote logic to bridge the two (logic and UI components). I have a few warnings on my activity_inventory.xml I may want to reevaluate.

I learned new tricks with how to incorporate android design that didn't require creating new xml files (pages, view windows, or additional screens which can lead to latency issues as the application grows). Utilizing view control (visibility and gone implementation) and search view allowed functionality on the same screen for better UI design. That was easily accessible and keeping the application simple.

The process of enhancing this artifact was difficult when incorporating all elements of the search view to my project. I had to create an onQueryTextSubmit to allow input and

onQueryTextChanged to take the user input and search the database and based on those findings clear the current inventory list and add the found item to a new filtered list to display. I needed to understand the sequence of utilizing the Search View. I faced challenges on the onCloseClickListener() I attempted to utilize this when the user exits out of the search view the screen would revert back to displaying the inventory list (prior to searching) and hide the search bar from view and have the spinner (sort drop down feature) and the search icon (magnify glass) reappear. However, I could not get the search bar to close. It was reading the data from the onClickListener because the full inventory list would reappear, however the search bar would remain at the top of the screen and would not revert to a hidden view. I attempted various methods to force close this mechanism without luck considering it was not a menu view. I noticed if I kept the search icon (magnify glass) in view (not hidden, in prior attempts I hid this) I noticed clicking on the search icon it would close the search view. It seemed like odd behavior. I used this to my advantage and created another image button within the activity inventory XML to have an x icon. I placed this next to my search icon and hid it from view. I applied the same trick of hiding the UI components depending on user action. When the search view was expanded I hid the spinner and search icon and had the x image icon in view to the left of the search bar and created an onclick listener for the close search button (the x icon) instead of utilizing the onCloseClickListener this allowed me to close the search view, hide it, and have the prior UI components display along with the original inventory list. It bypassed all other issues with the previous listener and allowed more control over the applications